

**Espresso
Extraction:
Measurement
and Mastery**

Scott Rao

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The author has taken care in preparation of this book but assumes no responsibility for errors or inaccuracies.

Preface

In my 20 years in the coffee business, I've seen a mere handful of cafés provide all the conditions necessary to produce a great espresso. My goal in writing this book is to teach baristi what those conditions are, how to optimize them, and how to ensure the result is a great espresso.

This book is not written for the novice barista; at the minimum, a reader should have several months of espresso-making experience and a basic understanding of the technical aspects of espresso extraction. As well, I strongly recommend reading and understanding *The Professional Barista's Handbook*, my comprehensive guide to espresso preparation, before tackling this work. *The Professional Barista's Handbook* provides a technical foundation that lets you better take advantage of the information in this book.

Introduction

Consistently producing ideal espresso extractions is fiendishly difficult. Simply to get the opportunity to pull a great shot, a barista must optimize numerous variables. Once a barista addresses those variables, she still must skillfully execute several actions to produce an ideal extraction.

Although barista skill and a little luck sometimes combine to create a fabulous espresso, being able to consistently produce delicious coffee requires both skill and the use of quality tools. These tools include a coffee refractometer, well-made baskets, and a grinder with large, sharp burrs. Baristi who avoid using all of these tools often make excuses implying that a barista's skill is all that is needed to pull great espresso.

Many baristi are uncomfortable with measuring extraction or with implementing systems to ensure consistent results. They seem concerned that such ideas take power out of the hands of the barista, but they shouldn't be concerned. While the past few years have seen the invention of the coffee refractometer, VST baskets, and machines with programmable pressure profiling, those years have also seen an unprecedented increase in the status and respect accorded to baristi. The wise barista will realize that the more tools and technologies he has at his disposal, the more ability he will have to control the quality of his coffee, and the more respect he will ultimately receive. No matter how much baristi take advantage of tools and technology, there will always be a place for judgment and skill in espresso making.

What Is an Ideal Extraction?

Below, I define an ideal extraction based on parameters that are rapidly becoming the industry standards for espresso normale:

Dry Dose

The dry dose should weigh between 14 and 22 g and should approximate the dose a given basket is designed to hold (i.e., within 10% of the basket's intended dose).

Brewing Ratio

Shot weight should be approximately 1.5 to 2.0 times the weight of the dry dose. I consider lower ratios to be “ristretto” and higher ratios to be “lungo” espresso.

Temperature

Water temperature at the group head should be 195°F to 203°F (91°C to 95°C). Generally, the lower temperatures in that range should be used for higher brewing ratios (more lungo), and higher temperatures should be used when brewing with lower ratios (more ristretto). Coffee origin, processing, and roast degree also influence the water temperature the barista should use.

Time

Time elapsed from the moment a barista starts flow from the group until the barista stops the flow should be 25 to 40 seconds. Baristi using “flat” pressure profiles should aim for the low end of the time range. Those pulling shots with progressive preinfusion should aim for the middle of the range, and baristi producing extractions using pressure profiling machines should target the top end of the time range. (See <chap>Chapter 11, “Pressure Profiling.”</chap>)

Extraction

Baristi will usually find that extractions in the 19% to 20% range yield the best-tasting shots. Please note three exceptions to this recommended range:

1. Very large burr sets, when sharp, may produce fantastic shots at extractions greater than 20%. However, I do not recommend extractions greater than 20% for baristi who use most of today's popular espresso grinders.
2. Small or dull burrs may necessitate extractions below 19% to avoid excessive astringency and bitterness.
3. A barista seeking a clean, fruit-forward, bright shot may choose to extract in the 15% to 16% range (a.k.a. “the little hump”; see www.jimseven.com/2010/11/08/the-double-hump). Such shots emphasize fruitiness at the expense of caramels and balance and can be interesting and enjoyable as straight espresso. (Please note: Some experts argue that the little hump is merely an artifact of a common extraction calculation error. I am confident that the little hump is real and

not the result of an error.)

But What about Taste?

While reading this book, you may decide that my ultimate goal is to pull a shot at x extraction and y strength. It's not; my goal is to make the most delicious coffee I can. But while discussing what is delicious is devilishly difficult, describing the data that denote deliciousness is not. So I'll stick to explaining the espresso-making process, how to measure results, and how to make use of the measurement data. I'll leave it up to the reader to decide how to apply these ideas to suit his or her taste.

While I don't recommend relying solely on data, I also don't recommend that baristi rely on taste alone to decide how to make espresso. The problem with relying solely on taste is that many variables in the espresso-making process have nonlinear effects. These nonlinear effects make it nearly impossible to find the optimal set of adjustments through trial and error guided by taste alone.

For example, if a barista pulls a shot and does not like the flavor, he may try using a finer grind. If he decides he liked the first shot better, he will then revert to using a coarser grind. However, it's possible the barista should have made the grind even finer to optimize the coffee's flavor. How could that be? Let's say the first shot was a 17.0% extraction. Espresso is typically sour and unremarkable at 17.0%. The shot with the finer grind may have been 17.7% and perhaps marginally worse-tasting than the first shot. Had the barista made the grind finer again to create an 18.5% extraction, he would have noticed an increase in ripeness and caramels and a decrease in sourness. Such nonlinear changes in quality are the norm in coffee extraction. Therefore, a barista who is guided by taste may "dial in" a shot by fiddling with various parameters, but it's nearly impossible for him to know whether he has settled on the best possible set of parameters.

The best strategy for dialing in espresso is to taste every shot during the process of systematically adjusting settings. While dialing in, the barista should use a coffee refractometer to frequently record extraction data, get feedback about the effects of various techniques and settings, and learn the ways that different extraction levels correlate with flavor. Such an approach takes patience and discipline, but it steepens any barista's learning curve and teaches him how to dial in shots more quickly and precisely.

Overextraction and Underextraction

For simplicity throughout this book we assume a barista's goal is to produce a normale with a 19% to 20% extraction. The barista will find that anytime an extraction variable is suboptimal, it leads to a decrease in extraction. This means that, when pulling a normale, it's easy to underextract and difficult, if not impossible, to overextract. Therefore, our goal in adjusting the various parameters of espresso making will be to maximize extraction. The barista fortunate enough to find it easy to pull shots at 20% may intentionally choose settings that decrease extraction. For everyone else, I recommend seizing every opportunity to maximize extraction. (For the record, it is usually easy to yield 19% to 20% when pulling a lungo, but it may be impossible to reach 19% when pulling a very restricted ristretto. For such a ristretto, aiming for 15% to 16% may be the best strategy.)

This Chapter addresses pump pressure in machines that do not have pressure profiling capability. For programmable pressure profiling, please see Chapter 11, “Pressure Profiling.”

An espresso machine applies pressure to force water through a packed bed of finely ground coffee. Generally, as pressure increases, so does flow rate through the coffee bed. However, pressure also compresses the coffee bed. At some pressure in the range of 7 to 9 bar (measured at the group), the flow rate peaks because, beyond that pressure, additional compression inhibits flow more than the additional pressure promotes flow.

If a barista operates a machine with manually adjustable pressure and she gradually increases pressure, she will find that the flow rate increases continually until approximately 8 bar, at which point flow plateaus and then begins to decline with further increases in pressure. Eventually, at some higher pressure, the flow will trickle, or even stop, because the coffee bed has become too compressed for water to force its way through.

I recommend using the pump pressure that produces the highest flow rate. This “optimal” pressure will allow the use of the finest grind that permits the barista to produce a normale in a reasonable amount of time; finer grinding facilitates higher extraction yields.

The pump pressure indicated on almost every machine’s built-in pressure gauge measures pressure at the pump, not at the group. Pressure at the group will be a little lower; how much lower will vary from machine to machine.

How to Optimize Pump Pressure

If you have a Scace-2 or other group-head pressure gauge, I recommend using it for this procedure. When adjusting pump pressure for each set of shots, use the device in lieu of a blind filter. If you do not have a group-head pressure gauge, reference the machine’s built-in pressure gauge for each setting.

In this procedure you pull approximately four sets of three shots each. Prior to pulling each set of shots, insert a blind filter in the portafilter and set the pump’s pressure while it is running. Use exactly the same dose for each shot, measured to a resolution of 0.1 g. For each shot, start a timer when the pump is engaged, and terminate the shot at 30 seconds by pulling the cup away from the flow. Do not adjust the grind during this test. It is critical that you practice identical procedures for dosing, distributing, grooming, and tamping each shot. If a shot channels or has some other defect, discard it and try again.

Step 1. Set the pump to 7.5 bar if using the machine’s pressure gauge (or 7.0 if measuring at the group.) Pull a shot, and terminate it at 30 seconds. Record the shot weight. Repeat this procedure two more times, and average the weights of the three shots.

Step 2. Increase the pressure to 8.0 bar (or 7.5 at the group.) Repeat the procedure, and record and average the shot weights.

Step 3. Continue increasing the pump pressure by 0.5 bar per set of shots.

Step 4. Once the average recorded shot weight decreases in successive sets of shots, the test is complete. The optimal pump pressure is the one that produced the greatest average shot weight.

Step 5. (optional). After determining the optimal pressure, you can refine it by repeating the test using smaller incremental changes in pump pressure. For example, if the test indicated an optimal pressure of 9.0, retest at 8.8, 9.0, and 9.2, and choose the pressure that yielded the greatest average shot weight in 30 seconds.

Pump Pressure	Average Shot Weight (30 sec)
7.5	30
8.0	33
8.5	35
9.0	31

Based on the results in this hypothetical example, the barista should set the pump pressure to 8.5 bar or, better yet, perform Step 5 and run a second test using pressures of 8.2, 8.4, 8.6, and 8.8 bar.

Water Quality and Extraction

I will not cover the topic of water chemistry extensively, as I wrote about it in depth in both *The Professional Barista's Handbook* and *Everything But Espresso*. However, I address some advanced concepts here:

Scale

It is difficult to know with certainty whether your water will deposit scale in your espresso machine. The best way to be sure your machine is safe is to have your water analyzed for TDS, hardness, alkalinity, and pH and to enter those numbers into the Langelier Saturation Index (LSI) formula along with various temperatures. (You can find various LSI calculators on the internet.) The LSI will indicate the rate of scale deposition at different temperatures. Be aware that your municipality's water chemistry may change seasonally and the temperature in your espresso machine's boiler will fluctuate.

Softening

Use caution when softening water that has high bicarbonate content. Many espresso machine manufacturers recommend or require softening water for all machine installations, as this is the surest way to prevent scale. However, softening usually compromises flavor and decreases the body of espresso. (Machine manufacturers are more interested in preventing you from ruining their machine and generating a service call than in helping you produce the best-tasting coffee.) As well, if the source water is high in bicarbonate prior to softening, the softened water will increase extraction time, sometimes dramatically, forcing a barista to coarsen the grind, thus decreasing total extraction.¹ With such water it may be impossible to pull 19% normales. I recommend doing a careful cost-benefit analysis before choosing to soften your espresso water.

General Water Chemistry Guidelines

I recommend the following brewing-water standards:

Water Chemistry	
Parameter	Recommended Range
TDS	120–130 ppm (mg/L)
pH	7.0
Hardness	70–80 ppm
Alkalinity	50 ppm

Water within these ranges will usually offer the best extractions. Of course, depending on the chemistry of your untreated tap water, you may have to compromise a little on these numbers. Other than pH, which should be very close to neutral, missing these targets by up to 15% is acceptable.

Alkalinity

Water with high alkalinity tends to produce chalky-tasting coffee, while low-alkalinity water increases coffee’s brightness.

Alkalinity	Flavor
Too low	Bright, acidic
Just right	Balanced
Too high	Chalky, flat

I am usually reluctant to endorse particular products, but I do recommend VST baskets. Prior to the availability of VST baskets, no baskets on the market had precise, consistent hole size, area, and shape, and none were checked for quality on an individual basis, using an imaging system. As well, most baskets were not perfectly round and of consistent inner diameter. Misshapen baskets forced baristi to use, perhaps unknowingly, smaller-than-ideal tampers to prevent the tampers from getting stuck in baskets. (Imagine trying to fit a round peg into an oval hole.)

VST baskets have adequate total hole area for use with the larger doses that result from finger-swipe dosing. (Historically, baskets had less hole area because they were designed for use with shallower doses.) Most non-VST baskets are manufactured by mechanically punching out of the holes, leading to incomplete punches and misshapen holes, which result in inconsistent flow rates and extraction levels. I remember struggling years ago to find three baskets similar enough to produce consistent extractions on all three groups of my espresso machine; frequently it was not possible. To guarantee consistency, I recommend baristi use only baskets with precise, laser-cut holes verified using an imaging system.

Hole Area and Extraction

For a given basket diameter, the ratio of hole area to ground coffee weight determines potential extraction. Greater hole area provides less flow resistance, allowing the use of a finer grind or larger dose.

More hole area → decreased flow resistance → finer grind → more extraction

Less hole area → increased flow resistance → coarser grind → less extraction

Basket Design and Volumetric Dosing

Even great baristi, paying close attention, will often err by a few grams per shot, resulting in inconsistent brewing ratios. For many years, volumetric machines have had the potential to not only match but exceed the shot-to-shot accuracy of a good barista, while allowing the barista to multitask without sacrificing shot quality. However, poorly made baskets have always limited the consistency of espresso from volumetric machines.

If a barista using a typical, non-precision basket doses accurately, uses a tamper that fits that basket snugly, and uses a programmable volumetric machine, he may end up with consistent results in the cup. If the barista then swaps portafilters among group heads, changing baskets in the process (but still using the same model and brand of basket), the results in the cup may change dramatically due to differences in the baskets' shapes, sizes, or open hole area. Shot volume, flow rate, and brewing ratio will all change if the baskets have different open hole area. Equally important, if the second baskets' diameter is too wide for the tamper, side channeling will occur, causing changes in flow rate, extraction quality, and brewing ratio.

The introduction of precision baskets by VST has allowed volumetric machines to finally fulfill their potential. With precision baskets and a volumetric machine, a barista can be confident that if he

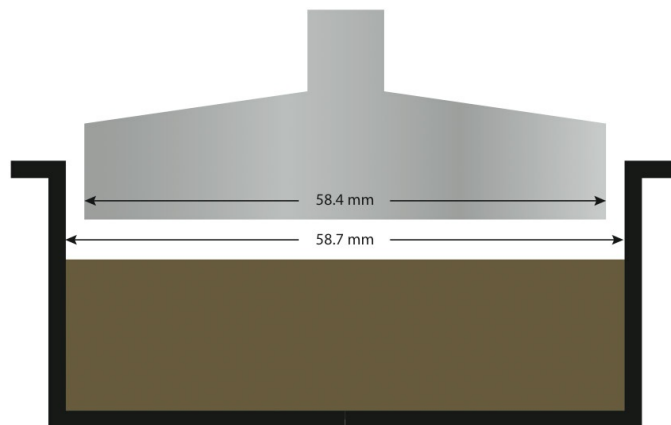
doses consistently and uses a properly fitting tamper, every shot will have nearly identical brewing ratios. He will only need to reprogram the machine when he changes basket size or brand.

Chapter 6

Tamper Fit

A tamper and portafilter basket must fit together snugly to minimize the risk of channeling. If a tamper's fit in a basket is too loose, channeling will frequently occur. Such channels may not always be visible, but diligent measurement with a coffee refractometer will show lower average extractions if a too-small tamper is used. Subtle channeling due to poor tamper fit can easily decrease extraction by 0.5 percentage point. The ideal tamper diameter is 0.25 to 0.30 mm smaller than the inner diameter of the basket. This translates to a gap of 0.125 to 0.15 mm between the tamper and the basket.

Tamper in Basket



In this drawing, my VST basket has an inner diameter of 58.7 mm and therefore requires the use of a 58.4-mm tamper, resulting in a gap of 0.15 mm around the perimeter of the tamper in the basket.

Roast Development

The internal structure of a coffee bean is a three-dimensional web of cellulose strands. Coating these strands is the soluble material that provides coffee's flavor. While the cellulose of raw coffee beans is "plasticky" and nonporous, roasting makes it more brittle and porous.² Under-roasted, or less developed, sections of cellulose are more plastic, resist fracturing during grinding, and are not porous enough for water to pass through. If water cannot penetrate a chamber enclosed by cellulose, the brewing water cannot extract the soluble material from that chamber.

Underdeveloped sections of beans result in lower total extraction as well as undesirable, savory flavors (See vstapps.com/blog-2/extractmojo/underdevelopment). Therefore, full roast development is necessary for both extraction quantity and quality. Unfortunately, with the recent (and generally welcome) trend toward very light roasting, few roasters consistently achieve full development. This fact (once dismissed as just my cranky opinion) is now verifiable with a coffee refractometer, as underdeveloped roasts yield lower extractions.

A roaster or barista may use the following procedure as a shortcut to evaluating roast development without pulling and measuring numerous shots:

Step 1. Boil water and grind coffee as you would to prepare a cupping.

Step 2. Allow the water to cool to 203°F to 204°F (95°C) before pouring.

Step 3. Pour the water over the grounds, and bring your nose to just above the surface of the brew immediately after pouring.

Step 4. Using a cupping spoon, break the crust of grounds and smell the aroma.

Step 5. Break the crust a few times and agitate the grounds while smelling, to get a firm impression of the aroma.

The more savory the aroma you detect, the less developed the roast. Aromatics reminiscent of broccoli, turnips, or celery usually indicate very poor development. Grass or straw are signs of more modest underdevelopment. This method is not foolproof, as some coffees may have intrinsic savoriness due to defects. Other coffees, such as Sumatras, may be herbaceous without being considered defective.

All else being equal, a roaster will improve development by roasting darker. I am not recommending darker roasting to improve development; however, sometimes roasters need to default to slightly darker roasting while figuring out how to properly develop a coffee at a lighter color.

A Few Notes on Underdevelopment

- Probably the most unusual case of underdevelopment I have witnessed involved a lightly roasted coffee that did not outgas at all after several weeks in a sealed bag. While this case was extreme, it did teach me how dramatically roast development can affect outgassing.
- I have frequently witnessed underdevelopment decrease extraction by 1 to 4 percentage points compared with the extraction yielded by a differently roasted batch of the same coffee.
- Subtle underdevelopment is perhaps the trickiest factor to decipher when trying to diagnose why

an espresso has underextracted. If an extraction is one-half of a percentage point below your expectation it is difficult to know whether the culprit is underdevelopment or some other factor, such as channeling. At the minimum, to rule out underextraction as the cause, the barista must measure multiple shots from the same roast batch. If some of them yield satisfactory extractions, then underdevelopment is not the problem. If all the shots are underextracted but a barista finds it easy to get more extraction from other roast batches, then underdevelopment is likely the problem.

As roasted coffee ages it releases gases, notably CO₂ and various volatile aromatics. Fresher beans contain more gas and release more gas during espresso brewing. Outgassing causes back pressure that inhibits the flow of liquid through the coffee bed. The additional outgassing of fresher beans decreases extraction by forcing the barista to use a coarser grind. This phenomenon may be a reason most baristi prefer to “rest” (a euphemism for “age”) beans for several days before using them to prepare espresso.

Another, more speculative, reason is that most baristi still terminate shots manually, based on a visual estimate of shot volume. Because fresher roasts release more CO₂ during extraction, the shots they produce have larger crema, and more volume relative to their weight. Few, if any, baristi fully compensate for the variation in shot volume caused by different degrees of roast freshness. For example if a barista pulls two 30-g shots, one using beans roasted yesterday and one from beans roasted 15 days ago, the older beans will yield a shot slightly larger than 30 mL, while the fresher beans’ shot volume will be almost 60 mL. How many baristi do you know who have the presence of mind and willingness to almost double a shot’s volume when using very fresh beans? Because of this issue, baristi who stop shots by volume effectively pull shorter shots when using fresher roasts. Therefore, they are unintentionally underextracting very fresh roasts.

To recap, a barista pulling shots with very a fresh roast must use a coarser grind, virtually guaranteeing underextraction. If that same barista stops shots by sight, he is likely to compound the underextraction problem by pulling the shots shorter than intended. The combined effects of these two dynamics cause many baristi to overestimate the flavor benefit of resting beans. While resting does improve flavor, much of the improvement baristi attribute to resting is really due to pulling longer shots with higher extraction as beans age.

Of course, resting beans is a double-edged sword. Along with loss of carbon dioxide, aging causes oxidation (unless one stores beans in an inert environment, such as in a vacuum-sealed, nitrogen-flushed bag) and loss of desirable aromatics. I recommend baristi attempt to optimize all other brewing parameters before relying on extensive aging to attain a desired extraction. Using coffee beans 2 to 3 weeks old can be acceptable, especially with light roasts, but should not be necessary.

A fresh, high-quality grind is the most important prerequisite for great espresso. When planning your espresso equipment budget, I recommend prioritizing the quality of your grinder over that of your espresso machine.

Pregrinding Versus Resting

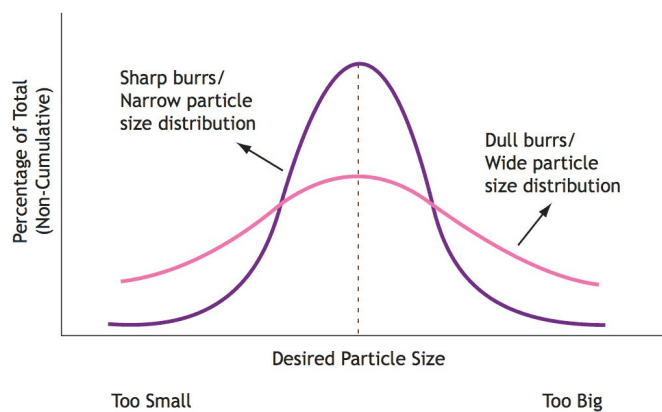
Grinding beans 30 to 60 minutes before pulling a shot yields results similar to those produced by resting beans for several days. (It is not feasible to translate the number of minutes ground to the equivalent number of days rested.) Despite this fact, most baristi frown upon pregrinding but embrace resting beans for several days or weeks. To be honest, I, too, prefer resting over pregrinding, but primarily because when practicing pregrinding in a café it is nearly impossible to control the amount of time between grinding and pulling a shot.

A final note on resting beans: Resting does not benefit nonpressurized brewing methods as it does espresso. In nonpressurized brewing most of the carbon dioxide released by the grounds dissipates into the air above the slurry. Outgassing causes turbulence in such brews but does not significantly affect extraction yield or flavor. I recommend using very freshly roasted beans for non-pressurized brewing.

Particle-Size Distribution

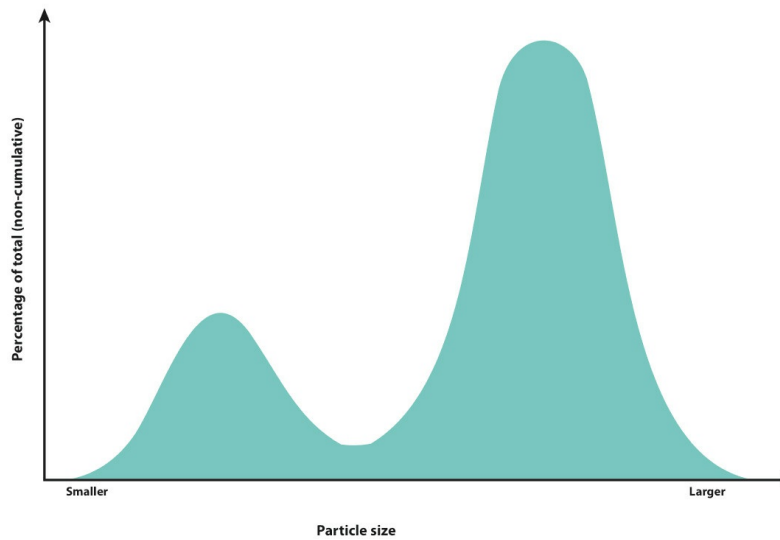
The most significant indicator of grind quality is particle-size distribution (PSD). Due to their irregular internal structure, coffee beans always fracture into a variety of shapes and sizes during grinding. A narrower PSD, meaning more of the grounds are close to the average particle size, indicates a higher-quality grind. A wider PSD contains a greater number of “fines” and “boulders.” Fines rapidly overextract, creating bitterness and astringency, while boulders tend to underextract, decreasing total extraction and brew strength.

DULL VS. SHARP GRINDER BURRS



This graph is an oversimplification designed to illustrate PSD. In reality, a PSD curve is not so smooth and does not have a single peak.

Coffee grinding, particularly at the fine settings used for espresso, produces a bimodal distribution³ like this:



The small peak represents the large number, but small mass, of fines, while the large peak indicates the large mass of grounds closer to the desired grind size.

Burr Size

Simply put, bigger burrs are better. Larger burr sets produce fewer fines and narrower PSDs, and they impart less flavor damage due to heat generation during grinding. Larger burrs are also essential in high-volume caf  s to prevent overheating during heavy use.

Because larger burrs produce narrower PSDs they yield better flavor and higher extractions than do smaller burrs. This effect can be dramatic; I've experienced some large-burr grinders capable of producing delicious 21% espresso extractions with minimal bitterness or astringency, while some grinders with small burrs cannot create espresso that tastes acceptable above 19.5% extraction. (I am comparing grinders having equally sharp burrs.)

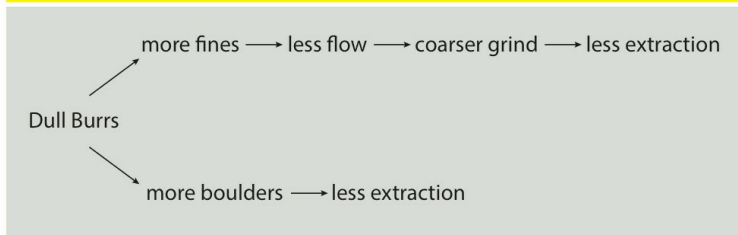
Not only do I prefer the better flavors and higher extractions from larger burrs, but higher yields can also save money. For example, a grinder producing 21% extractions decreases bean cost by 7% compared with a grinder that yields 19.5% extractions. If a caf   pays \$11 for a pound of beans, the savings would be approximately \$0.80 per pound.

Larger burrs cost more up front, but they dull more slowly, save money on beans, and produce better-tasting coffee. The extra initial cost of larger burrs is nearly always a worthwhile investment.

Burr Sharpness

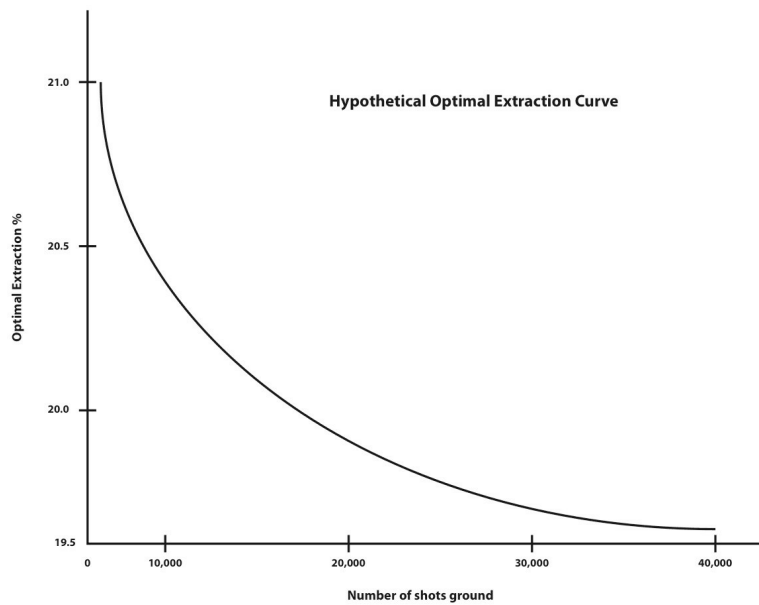
Sharper burrs produce narrower PSDs. As burrs dull, they create more fines and boulders.⁴ Additional fines increase surface area and decrease flow rate, and extra boulders decrease surface area and increase flow rate. One might think the simultaneous increase in fines and boulders would balance out each other's effects. Alas, this is never the case; the net results are always decreased flow rate and less extraction.

How dull burrs decrease extraction



Burr Sharpness and Optimal Extraction

Both the maximum and optimal extraction decrease as burrs dull. Therefore, it's too simplistic to say, "This Kenya tastes best at 19.5% extraction." It would be more accurate (but still simplistic) to say, "Using this sharp burr set, the Kenya tastes best at 20.5%, but 30,000 shots later, when the burrs are duller, it tastes best at 19.0%."



This graph represents the hypothetical optimal espresso extraction level (using a constant dose, coffee, roast, brewing ratio, etc.) for a grinder as its burrs dull.

Structural Grinder Problems

I've encountered three structural grinder problems that interfere with grind quality:

1. Divergent burrs: The most common problem I've encountered is non-parallel burrs. Typically, this has happened when a barista replaced the burrs after cleaning them. If even one or two ground particles remain under a burr while a barista is reseating it, those grounds can cause enough tilt in the burr to degrade grind quality.
2. Burrs not lined up with each other: Surprisingly, some professional grinders are not designed to force the burrs to line up perfectly. Reseating burrs in such a grinder requires the utmost care and a little luck. I've owned two such grinders, and both designs relied on the burrs' mounting screws to hold the burrs in place. Unfortunately, the holes in the burrs had larger diameter than the screws did, and a burr could easily be installed with an offset of 1 to 2 mm from its partner burr.
3. Bent motor shaft: I've encountered this problem with a small professional grinder. The shaft did

not spin “true,” which made the burrs not parallel. I never would have figured out the problem had not a friend with a machine shop measured the wobble in the shaft as it spun. I do not know how often this flaw occurs.

Doser Versus Doserless

In the 1990s, I rigged a funnel onto an espresso grinder to see if I could dose directly into a portafilter, circumventing the need to pull the dosing handle numerous times per shot. I succeeded in eliminating all the loud handle-pulling but found that the grounds in the portafilter basket were suddenly always clumpy. It dawned on me that the doser, probably unintentionally, declumped grounds.

To this day, I've yet to see a doserless grinder that does not create too much clumping (though every doserless-grinder manufacturer claims its grinders don't have this problem.) As well, every doserless grinder I've seen dispenses an unacceptably inconsistent amount of grounds per dose. For example, when I have attempted to pull shots with 18-g doses, the doserless grinders I've tested typically dispensed in a range of approximately 1.5 g (i.e., 17.25 to 18.75 g).

Most disappointing is the number of competitive baristi and respected caf  s that accept and use these wildly inconsistent doses. Inconsistent dosing guarantees erratic results, including fluctuating flow rates, extraction levels, brew strength, and flavor.

Allowing doses to vary by 1.5 g is unacceptable for a skilled barista. This problem with doserless grinders is not caused by their timers; the problem is the variable amount of grounds that gets stuck in the exit chute between the burrs and the dosing funnel. The grinder may in fact grind exactly 18.0 g per timer cycle, but the dose that falls from the chute will be 18.0 g plus (or minus) the decrease (or increase) in the amount of grounds remaining in the chute after grinding. An exit chute typically holds 4 to 5 g of coffee at any given moment, and that amount can vary by a couple of grams.

I recommend that you get rid of your grinder's doserless funnel and replace it with a traditional dosing chamber. With most, but not all, grinders it is easy to install a dosing chamber in place of the doserless funnel. Then use the following technique with a dosing chamber to make your dosing much more consistent:

Step 1. Prior to grinding, there should be 1 to 3 g of grounds in the dosing chamber.

(I respect the obvious protest that these 1 to 3 g will not be perfectly fresh. However, even when using a doserless grinder, an average of perhaps 4 to 5 g of preground coffee is trapped in the exit chute before each grind cycle. Another gram or more of residual grounds may be in the grinding chamber surrounding the burrs. Therefore, even baristi using doserless grinders typically pull shots containing 5 to 6 g of preground coffee. I would rather use very consistent doses containing 6 to 9 g of preground coffee than inconsistent doses containing 5 to 6 g of preground coffee.)

Step 2. Engage the timer to grind one dose.

Step 3. Pull the dosing handle repeatedly until the mound of grounds in the portafilter is approximately 1 gram more than the target dose. (This requires practice with a scale, but it is not difficult to master.)

Step 4. Allow the grounds to fall loosely into the center of the basket.

Step 5. Tap the portafilter on the counter several times to collapse any voids in the coffee bed. Be consistent in the number and intensity of taps you perform per shot. I recommend tapping the portafilter five or six times, as that will result in more-consistent doses than will tapping only once or twice.

Step 6. Using a Scottie Callaghan Dosing Tool or similar device, groom the surface of the coffee bed. It is essential to perform the same number of swipes each time you groom a dose. I recommend seven swipes in a N-S-E-W-N-S-N pattern, as shown:



Photo 1: The ungroomed mound of grounds.



Photo 2: Push the grounds “north” (i.e., to the far edge of the basket). Do not push any grounds over the edge of the basket.



Photo 3: Swipe south.



Photo 4: Swipe east.



Photo 5: Swipe west.



Photo 6: Swipe north, south, and then north again. Push any extra grounds over the edge of the basket after the final swipe north. This photo shows the remaining extra grounds before completing the final swipe.



Photo 7: The groomed bed of grounds.



Photo 8: Ensure the tamper is level and centered in the portafilter before applying firm pressure.



Photo 9: Tamp with 20 to 30 lb. of pressure.

Each swipe not only smooths the surface of the coffee bed but also settles the bed of grounds below the surface. Therefore, a greater number of swipes will tend to increase the weight of the dose, assuming there are “extra” grounds to push out of the basket on the final swipe. I’ve learned that standardizing the number of swipes makes the final dose weight more consistent. Seven swipes seems to strike a balance between efficiency and consistency.

Step 7. On the final swipe, push any extra grounds over the edge of the basket into the dosing chamber. Recycling this gram or so each shot will help maintain a constant 1 to 3 g of grounds in the dosing chamber.

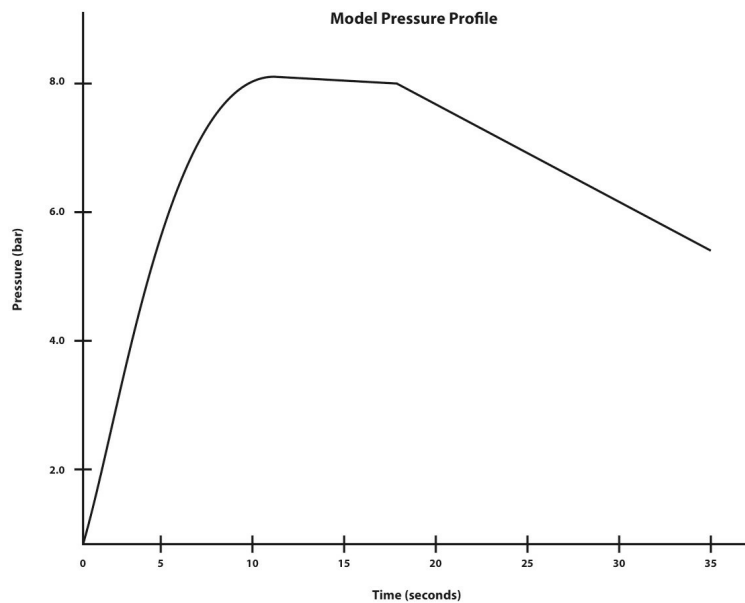
Step 8. Tamp once with modest pressure of 20 to 30 lb. (9 to 14 kg). Carefully remove the tamper without tilting it or disturbing the packed coffee bed. Do not tap the side of the portafilter to settle any loose grounds.

Step 9. Wipe the basket rim free of grounds before inserting the portafilter into the group head.

With practice, your doses should range no more than 0.4 g with this method (i.e., they will range between 17.8 to 18.2 g). If you want to achieve even better consistency, weigh each dose after grooming but before tamping, adjust the dose as needed, and groom again. To increase the dose, simply sprinkle more grounds into the basket by pulling the dosing handle fractionally, then regroom. To decrease the dose, regroom the coffee with a higher-numbered (more convex) dosing tool.

Using a scale provides feedback that will help you decrease your pregroomed dosing range. It helps to have a selection of dosing tools on hand to regroom after adjusting a dose.

Before I ever used a pressure profiling machine, my theoretical ideal profile consisted of a ramp-up preinfusion followed by a flat profile at a peak pressure near 8 bar and finishing with a lever-style linear decline in pressure. I confess to not having had as much hands-on experience with pressure profiling machines as I would like. However, I've advised several of my consulting clients to try versions of this profile, and so far no one has claimed to have improved upon this model. They have all adopted this profile with slight variations in their choices of pressures and times, and the shots I've had at their cafes have often been fantastic. I've gained two insights from my clients' experiences: (1) Optimal shot times are longer when using both a ramp-up and a declining finish, and (2) this profile facilitates higher extractions than does a traditional “flat” profile.



A friend of mine uses this profile to yield delicious espresso at extractions in the range of 19.5% to 20%. Note how the ramp-up of preinfusion is much steeper than the pressure decline in the second half of extraction. Also note the slight decline in pressure during the full-pressure phase. (This is explained below in “Quirks to Consider When Using Pressure Profiling.”)

Preinfusion (10 to 12 seconds)

Preinfusion is the initial wetting of the grounds at low pressure. The low pressure and slow flow of preinfusion allow the grounds to swell, redistribute themselves, and become more adhesive before full pressure is applied. The benefits of preinfusion include decreasing the frequency of channeling, limiting fines migration, and allowing the use of a finer grind. As I stated in *The Professional Barista's Handbook*, preinfusion will not necessarily make your best shot better, but it will almost certainly result in a much higher frequency of great shots.

Full Pressure (6 to 9 seconds)

Spending several seconds at the optimal full pressure maximizes total extraction. I recommend using the pressure that yields the greatest flow in a flat profile as a starting point and experimenting from there. (See “How to Optimize Pump Pressure” in <chap>Chapter 3.</chap>)

Declining Pressure (15 to 18 seconds)

Using declining pressure in the latter part of a shot may result in either more or less total extraction than would maintaining full pressure, depending on the rate of decline you choose. Allowing pressure to decline late in an extraction improves clarity, presumably by limiting the amount of fines that pass into the cup.

Quirks to Consider When Using Pressure Profiling

A ramp-up, ramp-down preinfusion program is unforgiving of any inconsistency in dry dose or tamping pressure. As with any volumetric machine, changing the dose by even a few tenths of a gram results in a substantially different brewing ratio. For example, if a barista increases the weight of the grounds from 18.0 g to 18.5 g, the brewed espresso weight will decrease by perhaps 3 to 4 g. The additional grounds absorb more water, preventing it from getting through the coffee bed, resulting in a lower brewing ratio. If you are going to use programmable pressure profiling, I recommend ensuring all doses are within 0.1 g of the target dose.

Tamping pressure influences head space above the grounds and preinfusion time. Increased tamping pressure has little effect on flow rate once full pressure is reached but delays the onset of full pressure, wreaking havoc on the machine's intended pressure curve. Decreasing tamping pressure shortens preinfusion time. While I don't think there is an "ideal" tamping pressure, if multiple baristi are going to share a pressure profiling machine, it is essential that they all calibrate their tamping pressures to a common standard.

When programming a pressure profile, you may notice that maintaining constant pump pressure in the middle of extraction results in a slow decline in pressure at the group head. This is normal, as the coffee bed provides decreasing resistance as it erodes. This phenomenon occurs in all machines, though a barista may not notice it until attempting to fine-tune a programmable pressure profile.

How to Use a Coffee Refractometer

A barista must zero-set a coffee refractometer each day before using it to ensure accurate, repeatable measurements. To zero-set a coffee refractometer:

Step 1. Clean the measurement glass, using an alcohol wipe.

Step 2. Rinse the glass with a few milliliters of distilled water.

Step 3. Wipe it dry with a clean, lint-free cloth.

Step 4. Put a few drops of distilled water on the measurement glass.

Step 5. Wait 30 to 60 seconds for the water temperature to equilibrate with the temperature of the glass.

Step 6. If you are using the Standard coffee refractometer, press and hold “CAL” for 5 seconds until “CAL” is displayed, and then simultaneously press and release “CAL” and “READ.” With the LAB coffee refractometer, put it in “SET ZERO” mode, and then press and hold “GO” for 5 seconds.

The temperature of the coffee sample should be within 1°C of the temperature of the distilled water when zero-set to ensure accurate readings. This is easiest to achieve if the user stores the coffee refractometer, distilled water, and sample cooling glasses near each other to ensure they are all the same temperature. It is also important that the temperature of the coffee refractometer is close to the ambient temperature where it is used. Should the coffee refractometer be warmer or cooler than the ambient temperature, allow the coffee refractometer’s temperature to settle to the ambient temperature before re-zeroing and measuring samples.

Measuring an Espresso Sample

Step 1. Record the weights of the dry grounds and espresso shot.

Step 2. Wait one minute after completion of the shot to allow CO₂ to diffuse out of solution.

Step 3. Draw 4- to 5-mL of espresso into a syringe from an area just below the crema.

Step 4. Attach syringe filter to syringe.

Step 5. Press the sample through the filter into a clean, dry, room-temperature cup.

Step 6. Allow the sample to cool for 30 seconds.

Step 7. Draw the sample into a fresh pipette.

Step 8. Transfer the sample onto the coffee refractometer measurement glass.

Step 9. Wait 30 seconds.

Step 10. Press “READ” or “GO” (depending on your coffee refractometer model) to take a measurement.

Step 11. Press the button several times in quick succession. The readings should vary no more than the stated precision range of your coffee refractometer. If the readings fluctuate more than expected, the sample may not have been properly filtered, or the glass not properly cleaned. Clean the glass, re-zero if necessary, and repeat the measurement process.

I recommend re-zeroing at least once per 30-minute period, in case of changes in room temperature.

How to Convert %TDS to Extraction Percentage

The most accurate way to determine extraction percentage is to input your %TDS reading, ground coffee weight, and brewed coffee weight (or brewing water weight) into VST's Coffee Tools™ software products. It is difficult to calculate extraction accurately without the software because one must take into account numerous factors, including the liquid retained in the grounds after brewing (an amount that varies depending on the brewing method used) and the likely moisture and CO2 content of the dry grounds. As well, if you are measuring liquid by volume, the software facilitates the temperature-dependent conversion of liquid volume to weight and the conversion between metric and English measurement systems. The software brilliantly takes into account all of these factors and produces simple, accurate calculations.

If you don't have access to the software, you may estimate the extraction percentage using the following formulae:

For percolation methods, including espresso:

$$\frac{\%TDS \times \text{Brewed Coffee Weight}}{\text{Ground Coffee Weight}} = \text{Extraction Percentage}$$

For infusion methods:

$$\frac{\%TDS \times \text{Brewing Water Weight}}{\text{Ground Coffee Weight}} = \text{Extraction Percentage}$$

Please note: The results of using these formulae may vary slightly from the software-generated results, depending on preferences used for CO2 content, moisture content, and liquid-retained ratio.

Using the Universal Brewing Control Chart™

The Universal Brewing Control Chart™ offers an elegant visual description of the relationships among taste, extraction, and brew strength. The charts generated by VST software are excellent training tools, helping baristi relate coffee taste and mouthfeel to the charts' various regions labeled "strong," "weak," "bitter," etc. The charts also allow baristi to compare results to traditional target regions, such as those of the SCAA or SCAE, or to a customized region, such as one tailored to the use of a high-quality grinder with a higher extraction target range. Most importantly, the "universal" nature of VST charts provide normalized extraction yield results regardless of the brewing method or brewing variables used.

Espresso

Recipes

Presets

Size

Single

7.0 g

Double

14.0 g

Triple

21.0 g

Concentration

Lungo

35.90% EBF

7.00 %

Normale

51.28% EBF

10.00 %

Ristretto

71.79% EBF

14.00 %

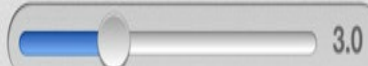
Parameters

RESET

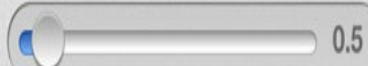
Extraction Yield



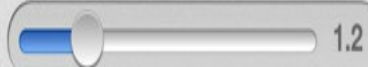
Moisture



CO2

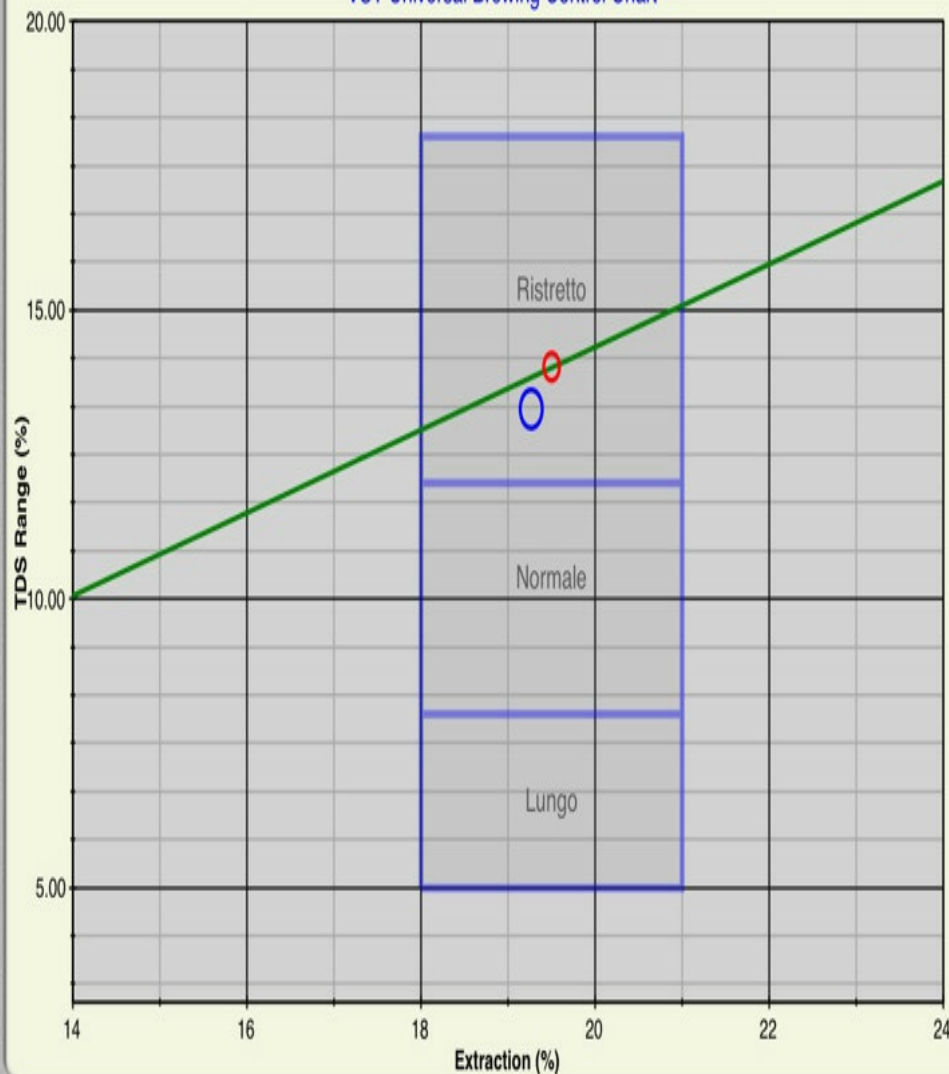


Liquid Retained Ratio



Apply

VST Universal Brewing Control Chart™



DESIGN

% EXT

EXTRACTION

19.50

19.4

19.5

19.6

% TDS

TOTAL DISSOLVED SOLIDS

14.00

13.9

14.0

14.1

DOSE (g)

DOSE WEIGHT

21.0

20

21

22

BW (g)

BREW WATER WEIGHT

48.0

47

48

49

BEV (g)

BEVERAGE WEIGHT

28.2

27

28

29



DRIP

PARAM	DESIGN	MEASURE
MOIST	3.0 %	EXT 19.50 %
CO2	0.5 %	TDS 14.00 %
	BEV	28 g
LRR	1.20	BEV 29 g
	EBF	74.40
	EBF	71.43
STARTING VALUE		
DOSE		

% TDS	MEASURED TDS
13.28	13.28
	13.2
	13.3
	13.4

BEV (g)	MEASURED BEV
29.4	29.4
	29
	30

This chart illustrates the extraction results of a very strong triple ristretto. The barista used a dose of 21.0 g to produce a 29.4-g shot with %TDS of 13.28% and extraction of 19.27%. The chart highlights that the shot fell into the range of “ristretto” and the %TDS and extraction were both slightly below the respective targets of 14.0% and 19.5%. Although the barista was probably pleased with the result, she might try making the grind a touch finer for her next shot.

Coffee

Recipes

Presets

Back



Edit

El Descanso

2013-05-12

extraction 20.00 %

total dissolved solids 1.45 %

dose weight 33.4 g

brew water weight 430 g

beverage weight 358 g

measured ext 20.42 %

measured tds 1.48 %

measured bev 341 g

score 9

brew method Immersion

brewer type Clever Dripper-Immersion

liquid retained ratio 2.50

moisture 3.0 %

co2 2.0 %

brew water temp 93°C

brew time 5 mins 45 secs

roaster name 49th Parallel

roast style Light

farm/farmer Carmen Rita Coca

grinder Virtuo

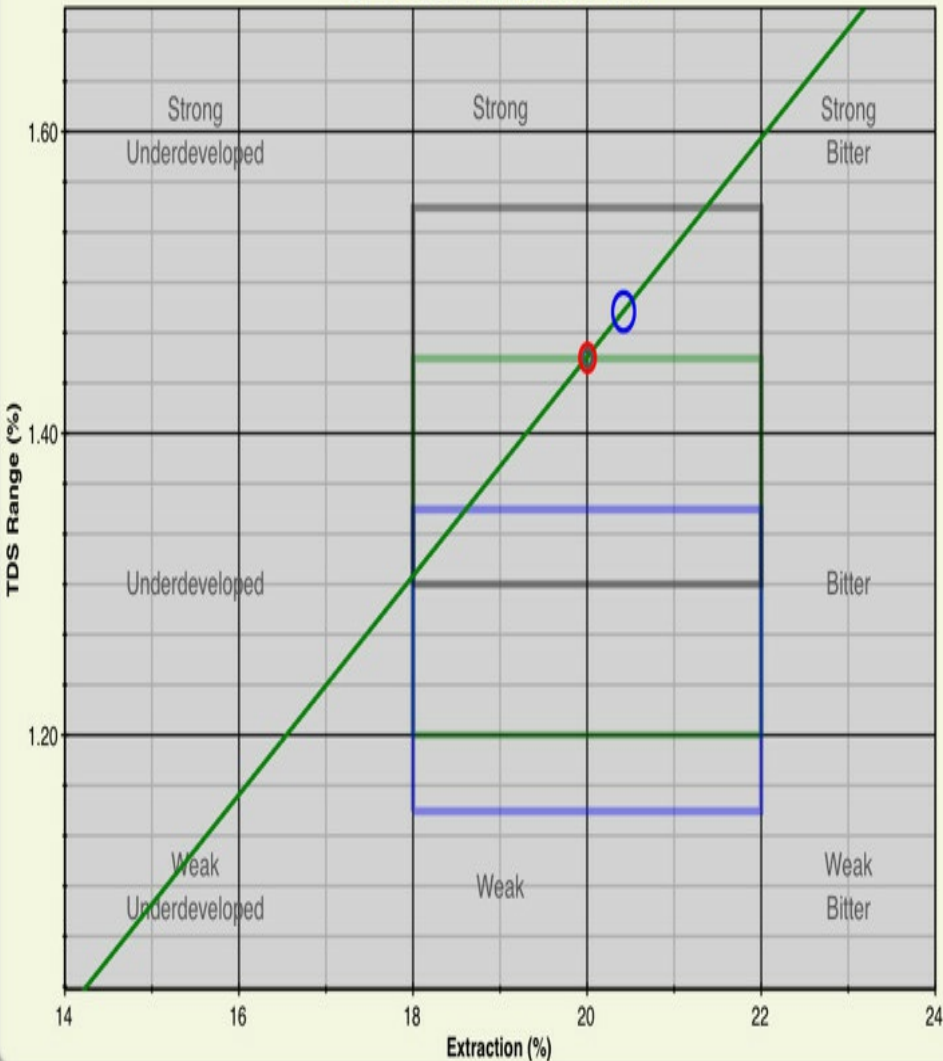
grind setting 23

roast date May 3, 2013

Sweet, light caramels, great clarity.
Full extraction at lightest possible roast.

Load Recipe

VST Universal Brewing Control Chart™



DESIGN

% EXT EXTRACTION

20.00

19.9 20.0 20.1

% TDS TOTAL DISSOLVED SOLIDS

1.45

1.4 1.5

DOSE (g) DOSE WEIGHT

33.4

33 34

BW (g) BREW WATER WEIGHT

430

420 430 440

BEV (g) BEVERAGE WEIGHT

358

350 360 370

DRIP IMMERSION

PARAM	DESIGN	MEASURE
MOIST	3.0 %	EXT 20.42 %
CO2	2.0 %	TDS 1.48 %
BW TEMP	93 °C	BEV 341 g
LRR	2.50	SPENT 122 g
STARTING VALUE		LRR 3.04
BREW WATER		

% TDS MEASURED TDS

1.48

1.4 1.5 1.6

BEV (g) MEASURED BEV

341

330 340 350

This chart illustrates the recipe parameters for “El Descanso,” roasted by 49th Parallel Roasters, brewed in a Clever Dripper with a long infusion, which categorizes the brew as immersion. This brew had a higher-than-intended Liquid Retained Ratio (3.04 versus 2.5) and extraction (20.42% versus a goal of 20.0%). The barista may consider grinding a little coarser on the next brew, although he also might keep the grind setting where it is, given that his notes indicate a very pleasing result in the cup.

The Benefits of Using a Coffee Refractometer

A coffee refractometer may have only one function—measuring %TDS—but that one data point offers myriad benefits. Knowing a brew’s %TDS can help a barista adjust brew strength, extraction, grind setting, and brewing ratio; estimate burr sharpness; evaluate barista consistency and roast development; and gain knowledge by way of experimentation. As noted, it can also help a coffee-taster learn to relate taste defects to different regions of the coffee-brewing control chart.

Brew Strength

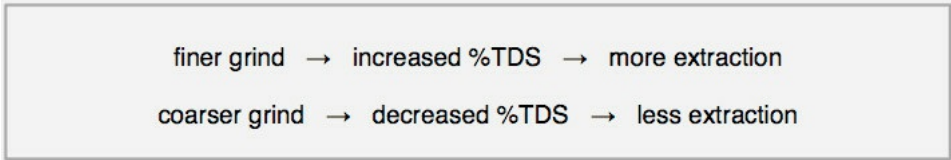
%TDS is a direct measure of brew strength. Ideally, a barista should have predetermined brew-strength target ranges for all coffees and brewing methods. Most professionals seem to prefer espresso brew strength in the range of 10% to 13% and nonpressurized brews between 1.25% and 1.45%.

Extraction

As discussed in <chap>Chapter 12, “How to Use a Coffee Refractometer,” calculating extraction requires knowing a brew’s %TDS. Perhaps the most valuable benefit offered by a coffee refractometer is the opportunity to compare the tastes of various brews with their extraction levels. </chap>

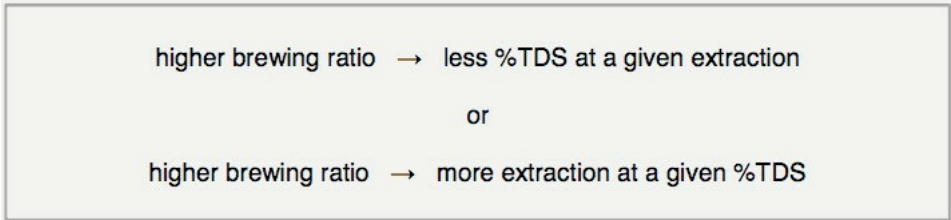
Grind Setting

Once a barista has measured %TDS and calculated extraction, if the numbers and taste are not to her liking, she can usually tune them by adjusting the grind.



Brewing Ratio

If a barista would like to change the resulting %TDS or extraction, but not both, she must adjust the brewing ratio.



If this idea is difficult to grasp right away, consider: As the barista runs extra water through espresso grounds (i.e., increases the brewing ratio), extraction increases, but brew strength decreases as the shot becomes a more dilute.

Burr Sharpness

Once a barista becomes familiar with a particular grinder, she can use a coffee refractometer to estimate the age of the grinder's burrs. To do this requires a fair amount of experience and the confidence that all other significant espresso-making variables are accounted for and well adjusted.

I recommend that every café record weekly extraction data for espresso and other brewing methods. At the minimum, the barista should record dry dose, shot weight, extraction time, %TDS, extraction percentage, and tasting notes, including the detection of bitterness or astringency. If the data are listed in spreadsheet format, trends in the data will indicate the condition of the burrs. The expected trend is for extraction and %TDS to slowly decline over several months while bitterness and astringency increase.

Barista Consistency

Logging such weekly data provides a fantastic objective indicator of barista consistency. Especially with manual, non-espresso brewing methods, I've found that rotating which barista on staff prepares the coffee to be measured always leads to some surprises. Most commonly, I've found that when baristi drift in their coffee-making techniques over time, they don't necessarily notice the impact of those changes on the coffee, but the coffee refractometer usually hints that something has changed.

Roast Development

Consistently low extractions with a particular roast batch often indicate insufficient roast development. While a barista may be able to diagnose underdevelopment by taste, with a coffee refractometer he can make that determination more confidently.

Education and Experimentation

Using a coffee refractometer allows a barista to test the effects of various techniques on extraction. For example, she could run tests comparing the results of using different-sized tampers, different water chemistries, or various distribution and grooming methods. If a barista runs a test measuring dozens of shots using each variation (i.e., using a 58.0-mm tamper versus using a 58.4-mm tamper), she will learn that variable's average effect on extraction level.

Firsthand knowledge of the ways in which varying dozens of parameters and techniques affects extraction is invaluable. A barista who has that kind of empirical data is able to diagnose extraction problems and improve results more easily.

For example, a friend of mine has a commercial espresso machine and grinder in his kitchen. (Actually, he has several. I have no idea how he has room to cook.) Frustrated by chronic underextraction, he embarked on some extensive data logging, using his coffee refractometer. After spending weeks toying with the variables, he found he could increase his average extraction by up to 0.5 percentage point by making any one of these changes:

- Changing his tamper size from 58.0 to 58.4 mm
- Replacing his old burrs with a new, sharp set
- Using E61-style preinfusion
- Using a different water source
- Switching to a darker or better-developed roast

This friend is brilliant and very skilled at making espresso, but I am not sure how he could have learned the effects of all of these changes had he not used a coffee refractometer.

Final Words

About 17 years ago I tried to convince my customers for wholesale roasted beans to grind each shot of espresso to order. Every client told me his shop was too busy to grind each dose separately. Eventually, almost all of them came around and decided to grind shots to order. I encountered the same resistance, and eventual compliance, when recommending that clients time each shot, use volumetric shot dosing, and use thermometers to verify milk temperature.

In more recent years, when VST introduced the coffee refractometer, many baristi felt threatened by the idea of measuring extraction, and they rejected the coffee refractometer as unnecessary. Many respected companies proudly scorned the coffee refractometer during the first few years of its existence. Thankfully, almost all of the critics have changed their minds and embraced Vince's coffee refractometers, software, and now VST baskets.

My hope in writing this book is to convince baristi of the benefits of measuring extractions and approaching espresso making systematically and scientifically. The future will bring more innovations that improve our ability to measure our results and consistently produce brilliant espresso. I hope baristi will evaluate these scientifically, with open minds. And I hope they remember that no matter what new tools are invented, those tools will complement, not replace, barista skill and judgment.

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Contact

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